CALCULUS · UNIT 2: LIMITS AND CONTINUITY

Toll Road Pricing

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design pricing for a toll road and garage. Step-function garage prices show jump discontinuities and one-sided limits, toll plans are made continuous by solving for parameters, limits at infinity describe average cost, and traffic counts use the IVT.

Fairness to drivers gives continuity a concrete meaning.

Standards Addressed

- AP LIM-1 Limits: graphical, numerical, algebraic, one-sided
- AP LIM-2 Continuity and types of discontinuities
- AP LIM-2.D Limits at infinity and asymptotes
- AP FUN-1.A Intermediate Value Theorem

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a parking receipt. Part 1 limits and jumps.
Day 2	Part 2 continuity and fixing tolls.
Day 3	Part 3 limits at infinity, IVT, and proposal.

Materials

- Student handout and Pricing Data Cards
- Graphing calculators or Desmos
- Optional: real toll schedules

How the Three Versions Differ

Level	What changes
CP2	Tables and graphs, one-sided limits at jumps, factoring a limit, a parameter for continuity, and an IVT explanation.
CP1	Classifying discontinuities, quadratic continuation, limits at infinity, and IVT with an equation.
Honors	Ceiling function, conjugate and squeeze limits, smooth piecewise design, and multiple IVT conclusions.

Watch For (Common Misconceptions)

- Assuming the limit equals the function value.
- Saying a limit does not exist when it is ∞ (describe it as unbounded).
- Applying the IVT to a discontinuous function.

Extension Ideas

- Investigate real parking rates in Boston.
- Graph the plans in Desmos with sliders for k.
- Write a letter to the authority defending your plan.

Answer Key – CP2

Part 1: Garage Prices

#	Task	Answer
1	Make a table of the price for 0.5, 1, 1.5, 2, and 2.5 hours.	$3, $3, $6, $6, $9
2	Find the limit of the price as $t \to 1^-$ and as $t \to 1^+$.	$3 and $6
3	Does lim_{t→1} P(t) exist? Explain.	No; the one-sided limits are different.
4	Is the garage price continuous at t = 1.5? Explain.	Yes; the price is $6 just before, at, and just after 1.5.

Part 2: Toll Plan A

#	Task	Answer
5	Find lim_{x→20⁻} T(x) and lim_{x→20⁺} T(x).	4 and 5
6	Is Toll Plan A continuous at x = 20? Why do drivers complain?	No; the toll jumps $1 at 20 miles.
7	Use a table to estimate $\lim_{x\to 3} \frac{x^2-9}{x-3}$.	Values approach 6
8	Find the same limit algebraically.	$\frac{(x-3)(x+3)}{x-3} = x+3 \to 6$

Part 3: Fix the Price

#	Task	Answer
9	Find k so that Plan B is continuous at x = 20.	4 = k + 1 → k = 3
10	Find the Plan B toll for 30 miles.	3 + 1.5 = $4.50
11	Traffic was 900 cars/h at 6 a.m. and 2,400 at 8 a.m. Explain why traffic must have been exactly 1,500 cars/h at some time in between.	Traffic changes continuously; 1,500 is between 900 and 2,400 (Intermediate Value Theorem).

Reflection

#	Prompt	Look for
R1	What does it mean for a price function to be continuous?	No jumps: the price just before and just after any point equals the price at that point.
R2	Why can a limit exist even if the function is not defined there?	The limit describes what the outputs approach, not the value at the point.

Answer Key – CP1

Part 1: Discontinuities

#	Task	Answer
1	Classify the discontinuities of the garage price P(t) for $0 < t \le 6$.	Jump discontinuities at t = 1, 2, 3, 4, 5
2	With the $20 daily maximum, write P(t) for $6 < t \le 24$.	P(t) = 20 (since 7 × 3 = 21 > 20)
3	Find $\lim_{x\to 3} \frac{x^2-9}{x-3}$ and classify the discontinuity of f at x = 3.	6; removable (a hole)
4	Redefine f to make it continuous at x = 3.	f(3) = 6

Part 2: Continuous Tolls

#	Task	Answer
5	Find k so that Plan B is continuous at x = 20. Show the three continuity conditions.	T(20) = k + 1; left limit 4; right limit k + 1 → k = 3; then T(20) = limit = 4
6	Design a toll $T(x) = ax^2 + b$ for $x \ge 20$ that is continuous at 20 and costs $8 at 40 miles.	400a + b = 4 and 1,600a + b = 8 → $a = \frac{1}{300}$, $b = \frac{8}{3}$
7	Find $\lim_{x\to 20}$ of your design from both sides to confirm.	Both sides approach 4
8	Is there a value of k that makes Plan A continuous? Explain.	No; Plan A is constant 5 for $x \ge 20$ and the left limit is 4.

Part 3: Pass vs. Pay

#	Task	Answer
9	Find $\lim_{n\to\infty}$ A(n) and interpret.	0.5; with many trips the average cost approaches $0.50
10	Find $\lim_{n\to 0^+}$ A(n) and interpret.	∞; with very few trips the pass is very expensive per trip
11	Use the IVT to show that the average cost equals $3 for some number of trips between 10 and 30. Then find it.	A(10) = 4.5, A(30) ≈ 1.83; 3 is between; 40 + 0.5n = 3n → n = 16

Reflection

#	Prompt	Look for
R1	What are the three conditions for continuity?	f(a) defined, lim f(x) exists, and lim f(x) = f(a).
R2	How do limits at infinity help compare plans?	They show the long-run average cost.

Answer Key – Honors

Part 1: Price Behavior

#	Task	Answer
1	Write the garage price using the ceiling function and find lim_{t→2⁻} P(t), lim_{t→2⁺} P(t), and P(2).	P(t) = 3⌈t⌉; 6, 9, 6
2	Find lim_{h→0} $\frac{\sqrt{(9+h)}-3}{h}$ and interpret it as a rate.	$\frac{1}{6}$; the instantaneous rate of √x at x = 9
3	A price-smoothing fee is g(x) = x² sin($\frac{1}{x}$) for x ≠ 0, g(0) = 0. Use the Squeeze Theorem to show g is continuous at 0.	−x² ≤ g(x) ≤ x² and both bounds → 0
4	Find lim_{x→∞} $\frac{3x^2-5x}{6x^2+7}$.	$\frac{1}{2}$

Part 2: Smooth Tolls

#	Task	Answer
5	Find a, b, c so that T is continuous at 20, has slope 0.1 at x = 20 (no sudden change in rate), and costs $10 at 60 miles.	At 20: 400a + 20b + c = 4; slope: 40a + b = 0.1; 3,600a + 60b + c = 10 → a = $\frac{1}{800}$, b = 0.05, c = 2.5
6	Verify continuity of Plan C at 20 with one-sided limits.	Left 4, right $\frac{400}{800}$ + 1 + 2.5 = 4 ✓
7	Why might a smooth price be fairer to drivers than a continuous one with a corner?	The price per extra mile does not jump.
8	Does Plan C ever charge less for a longer trip? Explain.	No; for x ≥ 20 the slope $\frac{x}{400}$ + 0.05 > 0, so it is increasing.

Part 3: Traffic & IVT

#	Task	Answer
9	Using the IVT, what is the minimum number of times traffic equals 2,000 cars/h between 6 and 9 a.m.?	At least 2 (once between 7 and 8, once between 8 and 9)
10	Could traffic have equaled 3,000 cars/h? Explain what the IVT does and does not say.	Possibly; the IVT guarantees values between data points but does not rule out others.
11	Write your pricing proposal with one discontinuous plan analyzed and one smooth plan designed.	Answers vary.

Reflection

#	Prompt	Look for
R1	What extra condition makes Plan C smooth and not just continuous?	The slopes of the two pieces match at x = 20.
R2	Why does the IVT require continuity?	Without continuity a function could jump over a value.